

MARCO FERNSTAEDT

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SOFTWARE DEVELOPER

PROFESSIONAL SUMMARY

Software developer with hands on project experience building full stack TypeScript applications, accessible AI tools, real time messaging systems, and collaborative coding platforms. Strong fit for junior software developer, frontend developer, web developer, and full stack developer roles requiring React, TypeScript, JavaScript, Next.js, Node.js, Express, MongoDB, REST API work, authentication flows, responsive user interfaces, accessibility awareness, and clear technical communication.

CORE SKILLS

JavaScript, TypeScript, React, Next.js, Vite, Node.js, Express, MongoDB, Mongoose, REST APIs, Socket.IO, WebSockets, HTML, CSS, Tailwind CSS, Git, GitHub, Clerk, JWT authentication, API integration, responsive design, accessibility, screen reader aware user interfaces, file upload handling, error states, debugging, documentation, customer communication.

PROJECT EXPERIENCE

InterviewLab Code Live Platform

Built a full stack live coding interview platform using React, TypeScript, Vite, Node.js, Express, MongoDB, Clerk authentication, Stream Video, Stream Chat, Monaco Editor, Inngest, and the Piston code execution API. Implemented authenticated interview sessions with host and participant roles, active and recent session tracking, and MongoDB backed session lifecycle management. Integrated video calls and chat channels per session so interviewers and candidates can communicate inside the coding workspace. Added browser based code execution for JavaScript, Python, and Java with visible output handling. Designed a resizable workspace with problem descriptions, starter code, editor, output console, video call, and live chat.

Socket IO Chat App

Built a full stack real time chat application using React, TypeScript, Node.js, Express, MongoDB, Mongoose, Socket.IO, Zustand, Axios, React Router, Tailwind CSS, and DaisyUI. Implemented authenticated WebSocket connections using JWT cookies and Socket.IO middleware. Added online presence, instant message delivery, optimistic user interface behavior, and persistent MongoDB message history. Integrated Cloudinary image uploads and in chat image previews for rich media messaging. Hardened backend flows with password hashing, protected routes, safe user serialization, centralized error handling, and Arcjet bot and rate limit protection.

Image Accessibility Tool

Developed a Next.js and TypeScript application that converts uploaded images into spoken MP3 descriptions for blind and low vision users. Built an API pipeline with multipart upload handling, MIME and file size validation, OpenAI Vision inference, text to speech generation, and binary audio streaming. Implemented accessible user interface patterns including screen reader live status updates, semantic controls, keyboard navigation, audio playback, and audio download. Added production abuse protection with Arcjet shield rules, bot detection, spoofed bot detection, hosting IP checks, and token bucket rate limiting. Handled browser Blob playback and download behavior, MP3 response headers, and image to text to audio workflow validation.

ADDITIONAL RELEVANT EXPERIENCE

Built and operated AI assisted workflows and automation systems for job search operations, reporting, call handling, inbox monitoring, and business process support. Comfortable reading application logs, source code, API behavior, user interface states, and browser errors to isolate defects and improve software behavior. Customer service background supports clear communication with users, support teams, developers, and hiring managers.

EDUCATION AND TRAINING

Self directed software development training through production style projects in React, TypeScript, Next.js, Node.js, MongoDB, authentication, APIs, real time systems, accessibility, and deployment workflows. CompTIA Network Plus in progress with expected completion in May 2026.